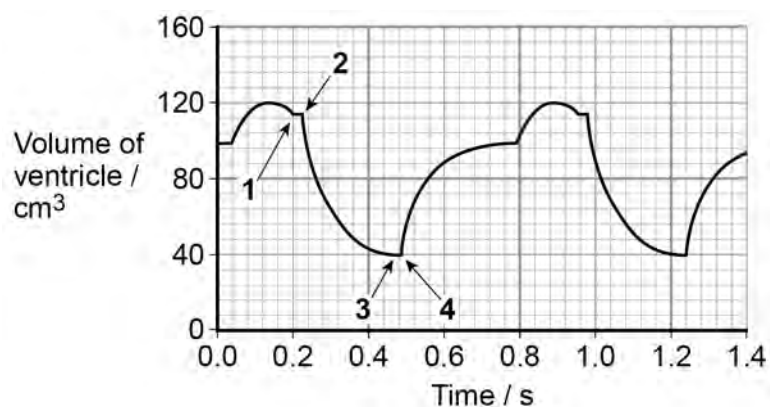


0 4

**Figure 3** shows the volume changes in the left ventricle of a human heart during two cardiac cycles. The numbers **1**, **2**, **3** and **4** represent times when heart valves open or close.

**Figure 3**



0 4 . 1

Use information from **Figure 3** to complete **Table 2**. Place the number **1**, **2**, **3** or **4** in the appropriate box.

[2 marks]

**Table 2**

|                               | Valve opens | Valve closes |
|-------------------------------|-------------|--------------|
| <b>Semi-lunar valve</b>       |             |              |
| <b>Atrioventricular valve</b> |             |              |

0 4 . 2

Use **Figure 3** to calculate the volume of blood pumped per minute by the left ventricle.

[2 marks]

Answer = \_\_\_\_\_  $\text{cm}^3 \text{ min}^{-1}$



0 4 . 3

Explain the role of the heart in the formation of tissue fluid.

**[2 marks]**


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0 4 . 4

Lymphoedema is a swelling in the legs which may be caused by a blockage in the lymphatic system.

Suggest how a blockage in the lymphatic system could cause lymphoedema.

**[1 mark]**


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7
**Turn over for the next question****Turn over ►**